Dipshikha Das

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Education

- **Atlantic Technological University, Galway** — Galway, Ireland
  PhD in Robotics & Manufacturing — *Oct 2024 – Oct 2028*
- **Tezpur University, Tezpur** — Assam, India
  B.Tech in Electronics & Communication Engineering; CGPA: 8.46 (Gold Medalist) — *Aug 2015 – June 2019*

Experience

- **Long Lab - Atlantic Technological University** — Galway, Ireland
  PhD Researcher - Robotics — *Oct 2024 - Present*
  - Conducting research on 3D Diffusion Policies for robotic manipulation, focusing on adapting and implementing the policy model for specific task execution.
  - Conducting research on robot learning and manipulation methods with a focus on exploring vision based robot learning using scene reconstructions approaches.
  - Using tools and frameworks such as ROS2, MoveIt2, perception pipelines, and policy learning libraries to design, test, and evaluate robotic systems.
  - Presented research poster on ongoing work (including winning contributions of Robothon Challenge) at IROS2025 late-breaking sessions and two workshops, highlighting primary findings and future directions.
- **Clutterbot** — Bengaluru, India
  Senior Engineering Manager - Robot Software — *Sept 2023 - Oct 2024*
  - Led the milestone growth of the robot software for indoor autonomous mobile robots from an MVP to a Beta stage in about 8 months, delivering 7+ core functionalities.
  - Engineered sophisticated decision-making algorithms and task planning for an autonomous robot system; having minimized human intervention by 80% and improved operational efficiency by 45% over six months.
  - Contributed to the design and implementation of a streamlined navigation and safety system for optimal robot performance by leveraging ROS2 and Nav2 features.
  - Collaborated with cross-functional teams to stabilize system requirements, ensuring seamless integration of robotics functionalities, leading to reduced integration time by 35%.
  - Led an adept team, driving R&D and strategic planning to develop and implement innovative autonomous system features, facilitating collaboration, and achieving project milestones.
  - Mentored a team of 12+ members, fostering professional growth by implementing agile and scrum strategies yielding an increased team efficiency and accelerated project delivery by 40%.
  - Worked with product managers and customer teams to define requirements and technical strategies, using data from real-world tests.
- **Addverb Technologies** — Noida, India
  Technology Lead - Mobile Robotics — *Nov 2021 - Sep 2023*
  - Led research, development, and maintenance of navigation modules for Autonomous Mobile Robots in industrial applications, enhancing the operational efficiency for varying payload capacity robots ranging from 100kg to 2500kg.
  - Directed the design, development, and testing of advanced navigation and controller algorithms using C++ and ROS/ROS2; significantly reduced path planning errors and increased robot efficiency by 30%.
  - Innovated multi-agent system features for simulation and real-world scenarios; optimized performance metrics, leading to decreased task completion time and higher operational reliability.
  - Guided cross-functional collaboration to define and stabilize system requirements for robotics functionalities, enhancing integration efficiency and reducing system downtime.
  - Optimized software system design in collaboration with engineering teams, significantly enhancing safety architecture and functionality for both hardware and software development.
  - Streamlined product backlog across perception, systems, & product teams, improving release cycles by 15%.
  - Led global expansion of AMR business through customer demos, on-site setups, and technical sessions with channel partners and achieved a 15% increase in international sales.
  - Managed, mentored, and trained a team of 3+ engineers to optimize code and algorithms using profilers and design principles, improving efficiency and performance by 20%.

- *Engineer - Mobile Robotics* *July 2020 - Oct 2021*
  - Engineered and expanded source code for multiple robot simulation tools, enhancing localization, sensor fusion, and navigation; revised performance metrics and stability by 35% through robust code development.
  - Diagnosed and rectified memory leaks using advanced debugging tools like Valgrind and GDB; improved code stability and performance by 60%, increasing overall module and system reliability.
  - Built and maintained internal systems and testing processes, reducing setup and testing time by 50%; trained junior engineers, enhancing team efficiency and knowledge transfer.
  - Administered innovative solution inputs with solution design and sales teams by comprehending customer requirements, allowing for faster outputs and simplifying the feedback cycle with improved performance.

- *Graduate Engineer Trainee - Mobile Robotics* *July 2019 - June 2020*
  - Conducted comprehensive literature surveys and spearheaded the development of advanced controllers for natural navigation-based robots, significantly enhancing navigation accuracy and reducing operational errors.
  - Executed feasibility studies, prototyping, and testing of 18+ robots, leading to successful deployment in industrial manufacturing plants and improving production efficiency.
  - Led end-to-end solution development and setup for warehousing exhibitions and customer demos, increasing customer engagement and securing two new client partnerships.

- *Intern - Mobile Robotics* *Jan 2019 - June 2019*
  - Collaborated with senior engineers to design and develop robotic systems for industrial automation focused on applying effective navigation and localization techniques, enhancing the combined accuracy.
  - Assessed and mastered varying aspects of robotics by engaging in learning opportunities for concepts of robot kinematics, dynamics, and control; strengthening theoretical concepts for practical applications.
  - Acquired hands-on experience by testing 10+ robots and calibrating sensors in all systems, improving integration efficiency by 20% through enhanced setup processes.

Projects

- **3D Diffusion Policy**: Setting up the 3D Diffusion Policy for robotic manipulation to adapt to case specific experiments and building required framework interfaces to integrate with existing systems. *Sep 2025*
- **Robothon Software Stack**: An end-to-end software framework for the Robothon Manipulation Challenge 2025 using Moveit Task Constructor (MTC), integrating perception, planning, and control to enable autonomous task configuration & execution. See Github *June 2025*
- **Customizable Docker Env**: A modular docker-based setup for using ROS/ROS2, with reproducible builds and simplified dependency management. See Github *Jan 2025*
- **Mecademic-ROS2 Integration**: The repository supports the usage of ROS2-Moveit2 for the Mecademic robot. See Github *Jan 2025*
- **Dockerized Grasping Repositories**: Containerized external grasping repositories by integrating with the custom docker env for easy setup and consistent deployments across systems. See Github *Jan 2025*

Skills

- **Technical**: C++, ROS/ROS2, Linux, Unit testing, Python, Bash, Finite State Machines, Behaviour Trees, Gazebo, StageROS, Moveit/Moveit2
- **Frameworks & Profilers**: REST API, MongoDB, Git, Docker, Cmake & Make, Valgrind, GDB, VSCode Debugger
- **Tools & Documentation**: Miro, Jira, Azure DevOps, Doxygen, Dia & PlantUML
- **Interpersonal**: Communication, Leadership, Problem-Solving, Teamwork, Empathy

Certifications

- **C++ Design Patterns** (Poash Technologies) *Apr – Jun 2023*
- **Agile and Scrum Methodologies** (Workshop by Shipra Aggarwal) *Mar 2023*
- **Structured Interviewing Skills and Training for Right Fit Hiring** (Jo Paulson) *Aug 2022*
- **Udacity Nanodegree in C++ Programming** (Udacity) *Apr - July 2020*

Accomplishments and Volunteering

- **Best Presentation Talk RISE@ATU Symposium**: Presented ongoing work as part of the joint symposium as part of the Innovation in Digital Manufacturing and Design (IDMD) PRTP. *Nov 2025*
- **Winner (Team Award) @ Robothon Grand Challenge 2025** : A Robotic Manipulation Challenge hosted as part of i_munich from TU-Munich, Germany *June 2025*
- **Winner (Irish Leg) & Qualified among top 7 teams @ Invent for the Planet 2025** : A 48-hour competition to build sustainable solutions for differenet challenges hosted by Texas A&M University College of Engineering *Feb 2025*
- **Volunteer @ Vex Robotics Competition**: Atlantic Technological University *Jan 2025*
- **Volunteer @ Teen-turn**: Atlantic Technological University *Nov 2024*
- **Recipient RISE@ATU**: Funded PhD Researcher under the TU RISE(Research and Innovation Supporting Enterprise) programme. *Oct 2024 - Present*
- **Top 50 @ Women In Robotics 2023:** Article *Oct 2023*
- **Meher Baba Competition, Indian Air Force**: Shortlisted for Phase 1(FOD detection - Swarm Drone) *Feb 2023*
- **Dream Team**: Voted as a member of Dream Team @ Addverb Technologies *July 2022*
- **Gold Medalist**: Awarded the Gold Medalist in B.Tech ECE Department @ Tezpur University *June 2019*
- **Chairperson & Member**: Led the IEEE Student Chapter @ Tezpur University *Aug 2018 - June 2019*